Chapter 15: Inference in Practice

The Practice of Statistics in the Life Sciences
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Part III: Foundations of Inference

Why Inference Requires Judgment

Chapter 14 introduced confidence intervals and hypothesis tests under simple conditions.

The formulas are only one part of a statistical analysis. Reliable conclusions also depend on how the data were collected, whether the model is reasonable and whether the numerical result matters in context.

Central Idea

Statistical methods are useful when their assumptions fit the study and their conclusions are interpreted with appropriate caution.

This chapter develops the practical judgment needed to use inference responsibly.

Chapter Roadmap

The chapter focuses on four practical questions.

1. When are confidence intervals and hypothesis tests trustworthy?
2. How do confidence level, variability and sample size affect a confidence interval?
3. What do statistical significance and a P-value establish, and what do they leave unanswered?
4. How can sample size be planned before data collection begins?

The first part of the chapter reviews the general theory. The second part applies these ideas to examples from science and public research.

The Simple Conditions for One-Sample z Procedures

The procedures introduced in Chapter 14 assumed that

- The observations came from a simple random sample
- The population distribution was Normal, or the sampling distribution of $\bar{x}$ was approximately Normal
- The population standard deviation σ was known

Under these conditions, the one-sample z procedures use

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}} \qquad \text{and} \qquad z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

The assumption that σ is known is uncommon in practice. Later chapters replace it with an estimate from the sample.

Where the Data Come From

Statistical inference relies on probability to describe how a statistic varies across possible samples or treatment assignments.

This reasoning is most reliable when chance enters the study design through

- **Random sampling**, which supports generalization from the sample to the population
- **Random assignment**, which helps balance lurking variables and supports causal conclusions

Practical Question

Before calculating an interval or a P-value, ask how the observations entered the study and which population or treatment comparison the design can support.

Random Sampling and Random Assignment

The two forms of randomization support different conclusions.

Design feature	**Main purpose**	**Conclusion supported**
Random sampling	Select a representative sample from a defined population	Generalization to the sampled population
Random assignment	Assign experimental units to treatments by chance	Comparison of treatment effects and stronger causal interpretation

A study may use one, both or neither form of randomization. The scope of the conclusion should match the design actually used.

Practical Problems After a Good Design

Even a carefully designed study can encounter difficulties after data collection begins.

- **Nonresponse** may make respondents systematically different from people who do not respond
- **Attrition** may create bias when participants who leave a study differ from those who remain
- **Measurement problems** may introduce systematic error that the sampling model does not describe
- **Complex sampling** may require methods that account for clustering, stratification or unequal selection probabilities

A standard formula cannot correct a fundamental flaw in the study design or a mismatch between the design and the analysis.

The Population Model

Parametric inference begins with a probability model for the population.

For one-sample inference about a mean, the model describes the distribution of the quantitative variable and identifies the unknown parameter μ.

The Normal condition matters because the procedure uses the sampling distribution of $\bar{x}$. The observations themselves do not need to form a perfect Normal distribution whenever the sample size is large enough for the Central Limit Theorem to apply.

Distinction

The shape of the population distribution and the shape of the sampling distribution of $\bar{x}$ are different ideas.

Sample Size and the Normal Approximation

The amount of non-Normality that a procedure can tolerate depends partly on the sample size.

- For a small sample, strong skewness or outliers can make Normal-based inference unreliable
- For a moderately sized sample, a roughly symmetric population is often sufficient
- For a large sample, the Central Limit Theorem makes the sampling distribution of $\bar{x}$ approximately Normal under broad conditions

There is no universal sample-size cutoff that works for every population. Stronger skewness and heavier tails generally require more observations.

Outliers and Nonresistant Statistics

The sample mean $\bar{x}$ and sample standard deviation s are sensitive to extreme observations.

An outlier can change the estimate, increase the standard error and alter the conclusion of a hypothesis test.

Before performing inference:

- Check for data-entry and measurement errors
- Examine a dotplot, histogram or another suitable graph
- Investigate scientifically plausible reasons for unusual observations
- Use an alternative method when a small sample contains severe skewness or influential outliers

An observation should not be removed only because its removal produces a preferred result.

A Practical Conditions Checklist

Before reporting an inferential result, consider the following questions.

1. What population or treatment comparison does the study address?
2. How were individuals selected and, if applicable, assigned to treatments?
3. Are observations sufficiently independent for the procedure?
4. Does the distributional model fit the data and sample size?
5. Are outliers, missing observations or measurement problems present?
6. Does the chosen procedure match the sampling or experimental design?

These questions determine whether the reported standard error, confidence interval and P-value have a defensible interpretation.

The General Form of a Confidence Interval

Many confidence intervals have the form

$$\text{estimate} \pm \text{margin of error}$$

The margin of error combines two quantities:

$$\text{margin of error} = \text{critical value} \times \text{standard error}$$

For a one-sample z confidence interval for a population mean,

$$m = z^* \frac{\sigma}{\sqrt{n}}$$

The estimate determines the centre of the interval. The margin of error determines its half-width.

Confidence and Precision

Two desirable features of an interval compete with one another.

- A **high confidence level** gives a procedure with a high long-run coverage rate
- A **small margin of error** gives a more precise estimate of the parameter

With the same data, a higher confidence level requires a larger critical value z^* and therefore a wider interval.

Confidence level	z^*	**Relative width**
90%	1.645	Narrower
95%	1.960	Intermediate
99%	2.576	Wider

Population Variability and Precision

The margin of error increases with the population standard deviation.

$$m = z^* \frac{\sigma}{\sqrt{n}}$$

- A smaller σ means that individual values are more tightly concentrated around the population mean
- A larger σ produces greater sample-to-sample variation in $\bar{x}$

Natural biological variation usually cannot be removed. Measurement error can sometimes be reduced by improving instruments, procedures or training.

The value of σ should describe the population and measurement process relevant to the planned study.

Sample Size and the Square-Root Relationship

For a fixed confidence level and population standard deviation,

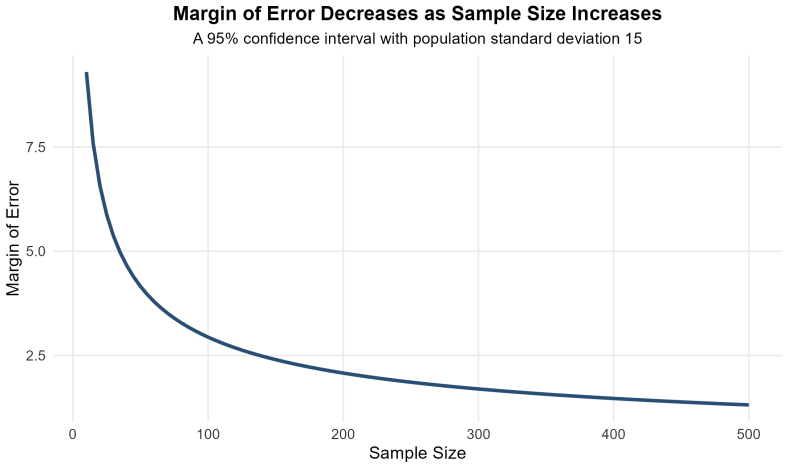
$$m \propto \frac{1}{\sqrt{n}}$$

Increasing the sample size improves precision, but the improvement follows a square-root relationship.

Change in sample size	New standard error	Effect on margin
$n \to 4n$	One-half as large	Margin is halved
$n \to 9n$	One-third as large	Margin is divided by 3
$n \to 100n$	One-tenth as large	Margin is divided by 10

Doubling the sample size reduces the margin by a factor of $1/\sqrt{2} \approx 0.707$, not by one-half.

Margin of Error and Sample Size



What the Margin of Error Measures

The margin of error describes uncertainty from random sampling under the assumed model.

It reflects how far the sample statistic may fall from the population parameter because a different random sample would give a different statistic.

Covered by the Margin of Error

Random sample-to-sample variation generated by the stated sampling process

A correctly calculated margin of error can still accompany a misleading conclusion if the observations are biased or the statistical model does not fit the study.

Errors Outside the Margin of Error

The usual margin of error does not measure uncertainty caused by

- Undercoverage or voluntary response
- Nonresponse and systematic attrition
- Leading questions or inaccurate self-reports
- Poorly calibrated instruments or inconsistent measurements
- Confounding in an observational study
- A statistical model that does not match the design or population

These sources of error can be larger than the reported sampling error. A narrow confidence interval does not compensate for biased data.

Evidence on a Continuum

A P-value measures how unusual the observed statistic would be under the null model, in the direction specified by the alternative hypothesis.

Smaller P-values provide stronger evidence against H_0, but there is no universal boundary separating meaningful evidence from unimportant evidence.

Interpretive Caution

The values $P = 0.048$ and $P = 0.052$ represent nearly the same strength of evidence, even though a fixed 5% decision rule places them on opposite sides of a threshold.

The exact P-value, study design and consequences of a decision should all inform the conclusion.

Type I and Type II Errors

A hypothesis-test decision can be correct or incorrect.

	H_0 **is true**	**A specified alternative is true**
Fail to reject H_0	Correct decision	Type II error
Reject H_0	Type I error	Correct decision

$$P(\text{Type I error}) = \alpha$$

$$P(\text{Type II error at a specified alternative}) = \beta$$

The relative seriousness of the two errors depends on the scientific and practical consequences.

The Alternative Hypothesis Determines the Tail

The alternative hypothesis must reflect the scientific question established before examining the sample result.

- $H_a : \mu > \mu_0$ uses the upper tail
- $H_a : \mu < \mu_0$ uses the lower tail
- $H_a : \mu \neq \mu_0$ uses both tails

For a symmetric test statistic, a one-sided P-value is often one-half of the corresponding two-sided P-value when the observed effect is in the stated direction.

Choosing the direction after seeing the data understates uncertainty and invalidates the intended error rate.

Statistical Significance and Sample Size

For a one-sample z test,

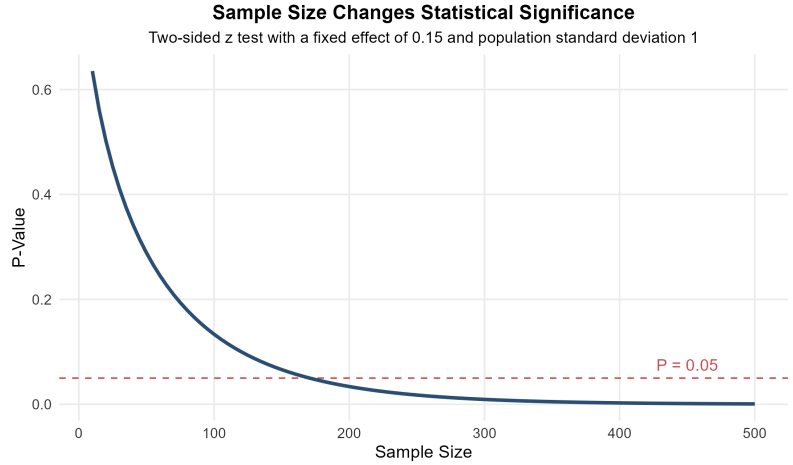
$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{\text{observed effect}}{\text{chance variation}}$$

The test statistic depends on

- The distance between the estimate and the null value
- The population variability σ
- The sample size n

A very large sample can make a small effect statistically significant. A small or highly variable sample may fail to detect an effect that matters in practice.

The Same Effect at Different Sample Sizes



Statistical Significance and Practical Importance

Statistical significance addresses whether the data are difficult to explain under the null model.

Practical importance addresses whether the estimated effect is large enough to matter in the scientific, clinical or policy setting.

Question	**Useful information**
Is there evidence of an effect?	P-value and model conditions
How large might the effect be?	Effect estimate and confidence interval
Does the effect matter?	Subject-matter thresholds, costs, benefits and consequences

A complete analysis reports the size and uncertainty of the effect, not only whether a threshold was crossed.

What a P-Value Does Not Mean

A P-value is calculated under the null hypothesis and the assumptions of the statistical model.

It is not

- The probability that H_0 is true
- The probability that the observed result occurred only by chance
- The size or importance of the observed effect
- Evidence that can be interpreted without checking the study design and model conditions

A large P-value means that the data remain compatible with the null model. It does not establish that the null hypothesis is correct.

Multiple Analyses and Selective Reporting

Each test conducted at level α has a chance of producing a Type I error when its null hypothesis is true.

Repeatedly testing many outcomes, subgroups or models increases the chance that at least one result will appear statistically significant by chance.

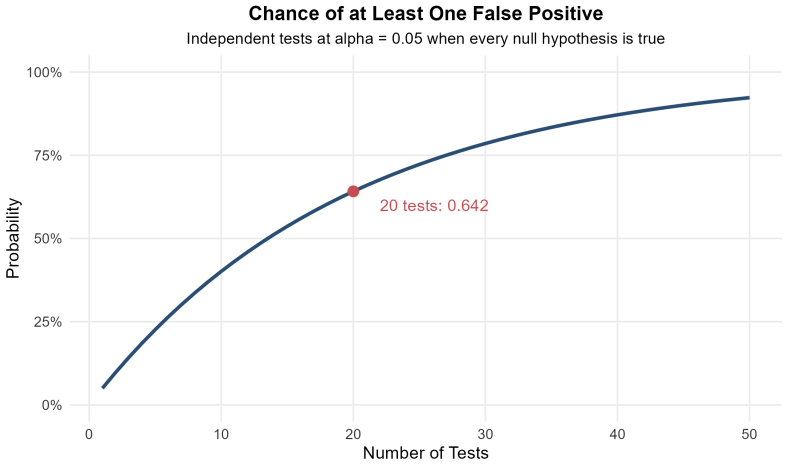
Illustration

If 20 independent tests are conducted at $\alpha = 0.05$ and all null hypotheses are true, then

$$P(\text{at least one rejection}) = 1 - (0.95)^{20} \approx 0.642$$

Searching until a small P-value appears is often called P-hacking or data dredging. Reporting only significant findings also creates publication bias.

False Positives Across Multiple Tests



Inference Within the Scientific Process

Statistical inference supports scientific reasoning when it forms part of a transparent process.

1. State the research question and analysis plan before examining the results
2. Use a design that connects the data to the intended population or causal question
3. Examine the data and assess the conditions for the procedure
4. Report estimates, uncertainty and exact P-values with appropriate context
5. Replicate important findings with new data when possible

Independent replication reduces the chance that a conclusion depends on an unusual sample, an unnoticed design problem or a flexible analysis choice.

Planning a Confidence Interval

For a one-sample z confidence interval,

$$m = z^* \frac{\sigma}{\sqrt{n}}$$

Solving for the sample size required to achieve a desired margin of error gives

$$n = \left(\frac{z^* \sigma}{m} \right)^2$$

The calculated value must be rounded **up** so that the achieved margin of error is no larger than the target.

The calculation requires a plausible value of σ, often obtained from previous studies or a pilot study.

Population Size and the Required Sample

The usual sample-size formula depends on

- The chosen confidence level through z^*
- The expected population variability through σ
- The desired precision through m

The population size does not appear in the formula when the population is much larger than the sample.

Interpretation

A sample of a particular size can estimate a mean with the same planned margin of error in two very large populations, provided that the sampling design and population variability are comparable.

Special corrections may be used when the sample is a substantial fraction of a finite population.

Planning a Hypothesis Test

Planning a hypothesis test requires three choices.

- **Significance level α:** the tolerated probability of a Type I error when H_0 is true
- **Effect size:** the smallest departure from the null value that would matter in context
- **Power:** the desired probability that the test will detect that effect

These choices should be made before data collection. They connect the statistical design to the scientific consequences of missing or incorrectly declaring an effect.

Power and the Probability of a Type II Error

The **power** of a test against a specified alternative value is the probability that the test rejects H_0 when that alternative value is true.

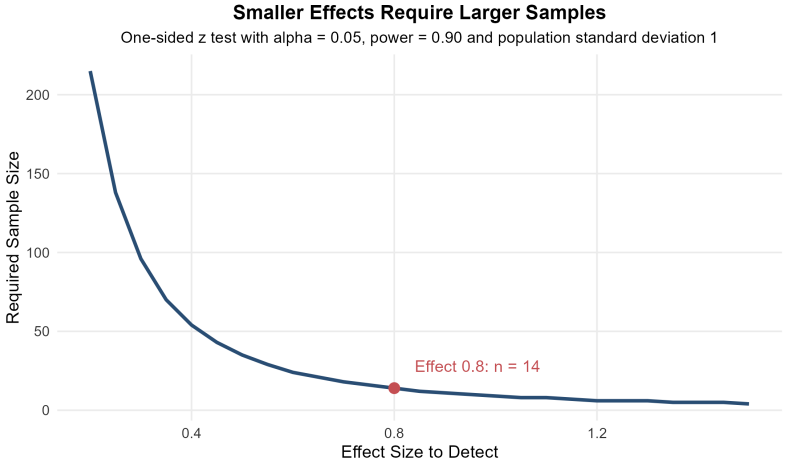
$$\text{Power} = P(\text{reject } H_0 \mid \text{specified alternative is true})$$

If β is the probability of a Type II error for that alternative, then

$$\text{Power} = 1 - \beta$$

Power generally increases with a larger sample, a larger effect, lower population variability or a less restrictive significance level. Two-sided tests generally require more observations than one-sided tests with the same target power.

Effect Size and Required Sample Size



Example 15.1: Two Convenience Samples

Two researchers recruit students enrolled in introductory courses at one university.

Neurobiology Study

A neurobiologist studies how people with normal vision respond to an optical illusion. Basic visual mechanisms may be similar across many groups, so subject-matter knowledge may support a cautiously broader interpretation.

Sociology Study

A sociologist studies attitudes toward using human participants in research. Age, education, field of study and campus culture may all influence the response, so one sociology class is unlikely to represent adults or university students generally.

Neither class is an actual random sample. The plausibility and scope of generalization differ because the research questions differ.

Example 15.2: Evaluating the Source of the Data

Three Chapter 14 datasets illustrate different levels of support for inference.

Study	Data source	Practical assessment
NHANES heights	Complex multistage probability sample	Broadly representative, but the formal analysis should account for the design
School-district IQ scores	Random sample of girls in the defined district	Supports inference to the stated district population
Blood phosphorus	Random records from one Ohio laboratory	Supports a narrow laboratory-related population, not all adults aged 75 to 79

The sampling frame and measurement setting determine the population to which the result can reasonably apply.

Example 15.3: Confidence Level and Margin of Error

For the IQ study,

$$\bar{x} = 105.84 \qquad \sigma = 15 \qquad n = 31$$

The margin of error changes with the confidence level.

Confidence	z^*	**Margin**	**Interval**
90%	1.645	4.43	(101.41, 110.27)
95%	1.960	5.28	(100.56, 111.12)
99%	2.576	6.94	(98.90, 112.78)

All three intervals have the same centre. Higher confidence produces a wider interval because it uses a larger critical value.

Comparing Confidence Levels



Example 15.3: Increasing the Sample Size

At 95% confidence, the original margin of error is

$$m_{31} = 1.960\frac{15}{\sqrt{31}} \approx 5.28$$

If the sample size increases from 31 to 62 while *sigma* remains 15, then

$$m_{62} = 1.960\frac{15}{\sqrt{62}} \approx 3.73$$

$$\frac{m_{62}}{m_{31}} = \sqrt{\frac{31}{62}} = \frac{1}{\sqrt{2}} \approx 0.707$$

Doubling the sample size reduces the margin by about 29.3%. A new sample would usually also produce a different value of $\bar{x}$, so the centre may change.

Example 15.4: A Large Social-Media Experiment

A randomized experiment altered the amount of emotional content shown in the News Feeds of 689,003 Facebook users for one week.

The reported tests produced very small P-values:

$$z = -4.78 \qquad \text{and} \qquad z = -7.219 \qquad P < 0.001$$

The estimated change was approximately one fewer emotional word per 1000 words written during the following week.

Interpretation

The large sample made the analysis sensitive to a very small average change. The small P-values provide strong evidence against a zero-effect model, but they do not show that the effect was large.

Example 15.5: Assessing Practical Importance

The Facebook experiment raises questions that a P-value cannot answer.

- Does using fewer emotional words reflect a meaningful change in emotion?
- Could users simply repeat words that recently appeared in their feeds?
- Would the experimental manipulation resemble ordinary use of the platform?
- Could a very small individual effect matter when applied across a large population?

Evaluating practical importance requires the estimated effect, its uncertainty, the quality of the measurement and knowledge of the setting.

Conclusion

Statistical significance answers a narrow probability question. Subject-matter reasoning determines whether the finding is meaningful.

Example 15.6: Planning a Body-Temperature Study

Researchers want to estimate the mean body temperature of healthy adults within 0.05°F at 95% confidence. Previous research suggests

$$\sigma = 0.6^{\circ}\mathrm{F} \qquad \text{and} \qquad z^* = 1.960$$

The required sample size is

$$n = \left(\frac{1.960(0.6)}{0.05}\right)^2 = 553.1904$$

The study must recruit at least

$$\boxed{n = 554}$$

Rounding down to 553 would produce a margin of error slightly larger than the target.

Example 15.7: Planning a Sweetness Study

A quality-control engineer studies the mean loss of sweetness after cola is stored. Each trained taster gives a before-and-after rating.

$$H_0 : \mu = 0 \qquad H_a : \mu > 0$$

The planning values are

- Population standard deviation $\sigma = 1$ point
- Significance level $\alpha = 0.05$
- Important mean loss $\mu = 0.8$ point
- Desired power 0.90

The initial design uses $n = 10$ tasters. The next step is to determine whether this sample provides the target power against $\mu = 0.8$.

Example 15.8: Calculating Power for $n = 10$

For an upper-tail test with $\alpha = 0.05$, the critical sample mean under H_0 is

$$\bar{x}_{\text{crit}} = 0 + 1.645\left(\frac{1}{\sqrt{10}}\right) \approx 0.520$$

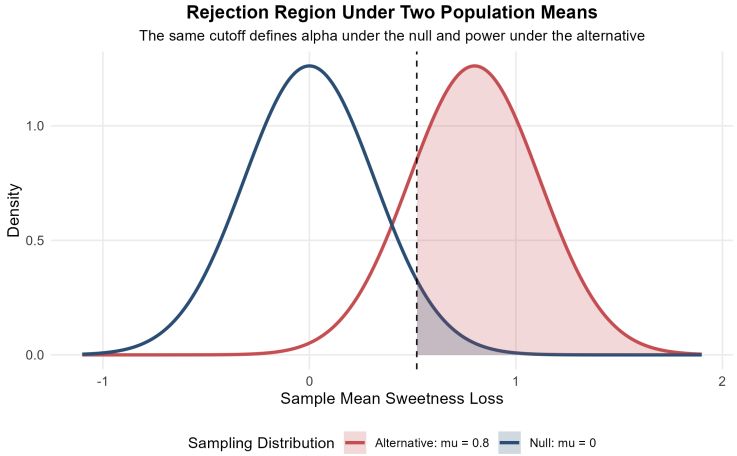
When the true mean loss is $\mu = 0.8$, the power is

$$P(\bar{X} \geq 0.520 \mid \mu = 0.8) = P\left(Z \geq \frac{0.520 - 0.8}{1/\sqrt{10}}\right)$$

$$= P(Z \geq -0.886) \approx 0.812$$

The test detects a true mean loss of 0.8 point about 81.2% of the time. Ten tasters do not provide the desired 90% power.

Null and Alternative Sampling Distributions



Example 15.9: Error Probabilities in the Sweetness Study

The test uses $\alpha = 0.05$ and has power 0.812 against the specific alternative $\mu = 0.8$.

Type I Error

$$P(\text{reject } H_0 \mid \mu = 0) = \alpha = 0.05$$

The study declares a sweetness loss even though the population mean loss is 0.

Type II Error at $\mu = 0.8$

$$\beta = 1 - \text{power} = 1 - 0.812 = 0.188$$

The study fails to detect a mean loss of 0.8 point when that loss is real.

Chapter 15 Summary

Reliable inference requires more than substituting numbers into a formula.

- The data-collection process determines the scope of generalization and causal interpretation
- Confidence intervals become wider with higher confidence or greater variability, and narrower with larger samples
- The margin of error measures random sampling variation rather than every possible source of error
- Statistical significance depends on effect size, variability and sample size
- A P-value does not measure the probability that H_0 is true or the practical importance of an effect
- Study planning connects sample size with precision, significance level, meaningful effect size and power

Clear conclusions report the estimate, uncertainty, study limitations and practical meaning together.